Take home notes for: Sauerkraut and Kimchi

During the workshop there can be a lot of new information to take in, don't worry! Here's a comprehensive guide to making sauerkraut and kimchi.

Base Recipe 1: Sauerkraut

Sauerkraut was the first vegetable fermentation I tried making and I always make enough to have a constant supply. It goes really well as a side with so many different dishes (see **Serving**).

The recipe and process from the first time I made it to now has changed very little. It's easy, quick, healthy, it makes cabbage last longer, and it's delicious. All you need is cabbage, salt and time. Any cabbage variety can work, even Brussels sprouts! But my favourite is just the bog-standard white or red cabbage.

Here's the recipe by ratios, so whatever amount of cabbage you have, you can always make it the same way.

Equipment

- Chopping board
- A knife
- Weighing scales
- Large bowl
- Small bowl
- A clean jar(s) (see **note on jars**)

Ingredients

- White or red cabbage
- Salt

Method

Firstly, take an outer leaf off the cabbage and put it to one side, more on this later.

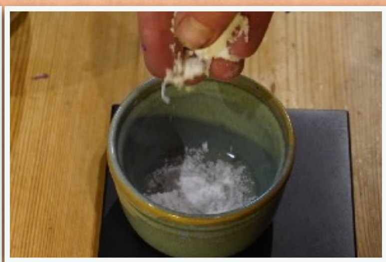


Take whatever amount of cabbage you have and slice it thinly. You don't have to be too precise but be aware that the chunkier the cuts, the longer it will take to ferment.



Add your chopped cabbage to a bowl with the scales zeroed underneath and weigh it.

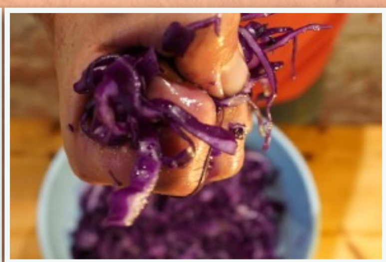




We need to add 2% of this weight in salt e.g. if your cabbage is 1000g, then you need 20g of salt. I find it easier to weigh the salt out in a separate small bowl, in case you add way too much by accident.. it’s happened before!

Now take your measured salt and add it to the bowl of chopped cabbage. Our aim now is to massage the salt into the cabbage. As you're doing this, scrunch the cabbage in your hands, we're trying to break down some of the cell walls so it releases the water it's holding. After a few minutes of doing this, cover





the bowl with a cloth to stop insects getting in and leave it for 30 minutes (or longer if you have other stuff to do). The salt will continue to draw moisture out of the cabbage.

When you come back to the bowl, you should notice the cabbage is wetter and a bit easier to scrunch.

Now transfer the cabbage to your jar(s), a handful at a time, it's going to be a bit messy, that's kind of unavoidable. As you add the cabbage to the jar, force it down with your hand, compacting it as you go.



As you fill the jar, the juices from the cabbage should start to rise up and cover the cabbage. If you're not seeing much juice, really press the cabbage down using force (hold the jar firmly with your other hand, it's glass, so you don't want it to slip and break).

Once the jar is almost full, use your saved cabbage leaf and fold it into a small enough parcel to stuff into the mouth of the jar, use it to push the cabbage



down and seal on the lid. There are other ways to weigh down the cabbage if you prefer (See **Why submerge the vegetables?**). You want the juices to cover the chopped cabbage once the lid is on, it doesn't need to cover your outer cabbage leaf parcel. Place your jar on a saucer.


Fermentation

Leave your jar of cabbage on the kitchen side at room temperature to transform into sauerkraut. This stage is very simple, just give it time. The longer it goes, the more sour it will get, but it will also become more umami. The right balance of these flavours is down to personal taste. To my palate, sauerkraut takes 2–3 weeks before it starts to hit its stride (depending on ambient room temperature), but while it's fermenting, open it up from time to time and give it a taste. As you make more, you'll get a feel for when it's not fermented enough and just tastes like salty cabbage and when it's gone too far and it’s really sour.

When it reaches a point you're happy with, simply put it in the fridge. This doesn't completely halt fermentation, but it does drastically slow it down, meaning the sauerkraut stays how you like it for much longer.

That's it! Enjoy.

Adding extra ingredients

This is a very minimal sauerkraut recipe, meant to help you get a handle on the process first. The possibilities for other additions are endless, and I'd highly encourage you to experiment.

A few of my favourites:

- **Grated carrot**, adds colour and sweetness to white cabbage sauerkraut
- **Chopped garlic**, adds pungency, don't overdo it though, fermentation amplifies its flavour
- **Caraway seeds**, give an earthy, slightly anise flavour
- **A bit of red cabbage** added to a white cabbage kraut makes it a vibrant pink colour

You could also try:

- **mustard seeds**
- **black pepper**
- **juniper**
- **dill**
- **chillies**
- **peppers**

Beyond these simple additions, this style of fermenting can be applied to basically any vegetable or fruit, and is a great way to turn leftover veg into tasty condiments. I often thinly chop cauliflower ribs and treat them exactly like cabbage for sauerkraut, or I've fermented carrots, courgettes and peppers together because that's what was in the fridge, it was delicious. Use your imagination and see what you can create.

Base Recipe 2: KimChi

Kimchi is often a crowd pleaser. It's bright red, tangy, spicy, pungent and packed with umami. It's a more complicated recipe than sauerkraut but it's worth it and it's ready quicker. Because of it's big flavour profile I find it pairs better with bold spicy dishes.

Kimchi in Korea is a preparation method rather a specific recipe. That is to say there are many different kimchi's using different ingredients and similar processes. I don't claim that this method I use is "authentic" but it is what works for me.

A note on amounts

With the amount of ingredients in kimchi, it makes using exact ratios somewhat cumbersome. I find a much better approach is to get a rough feel for the recipe and then with each batch you make try adding more of this or less of that, eventually you will find a balance you like. Here I am basing the amounts of things on one head of Chinese leaf cabbage, just try to get the amounts roughly, it will be fine. As long as your salt and soy sauce amounts are about right, all good.

Ingredients

The Vegetables:

- 1 large or 2 small heads of Chinese Leaf Cabbage (Napa Cabbage)
- 150g Mooli (Daikon Radish)
- 4 Spring Onions
- 1 Carrots
- Salt (2% total vegetable weight)

For the paste:

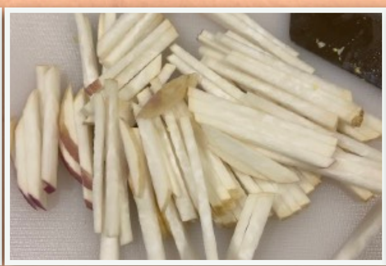
- 4 cloves Garlic
- 2 thumb sized pieces Ginger
- 1 apple, cored
- 60g Gochugaru (Korean Chili Flakes)
- 4 tbsp Soy-sauce
- 30g Rice Flour


Method

For the Vegetables:

The first part of this recipe is very similar to the above sauerkraut recipe.

Take your vegetables and chop. Try to think about the size of the pieces you'd like in the finished kimchi and chop to that. Normally I cut the cabbage into inch sized squares and the rest into very thin strips.


Then weight these vegetables and add 2% salt by weight. Do the same scrunching method as before but slightly less vigorous as Chinese leaf cabbage is more delicate.



Leave the vegetables in the bowl to one side whilst you make the paste.

Paste:

Add the rice flour to a pan and add 300ml water (1:10 ratio). Bring to a simmer and stir regularly so it doesn't stick to the pan. As it heats the mixture will gelatinise and thicken. Once you reach this you can take it off the heat and let it cool.

Whilst this cools, either grate or blend the garlic, ginger and apple. When the rice flour mixture is cool enough to touch add the ginger, garlic and apple and the Gochugaru and soy-sauce, stir to mix.

Add this mixture to your chopped vegetables and massage to coat them in it. Stuff everything into jars as described in Sauerkraut recipe, you can use your reserved cabbage leaf to submerge everything. Seal the lid and ferment.

Fermentation

This again is almost exactly the same as for sauerkraut although it does ferment much quicker so try it more regularly to see when it's done. In my experience 4 - 6 days is plenty. Move to the fridge when you are happy with the acidity levels.

Adding Extra Ingredients

As mentioned earlier, Kimchi is not one recipe but an approach to fermenting vegetables. This is very useful especially given that Chinese Leaf cabbage and Mooli can sometimes be difficult to get in super markets. You could substitute any cabbage here but I find a more tender cabbage, like a Sweetheart, works best. You could also substitute the Mooli for red radish or white turnip, the results will be very similar.

You can also get more creative. You could omit the spicy pepper flakes and make a white kimchi or try a version that has no cabbage but lots of radish. Sometimes I use Kimchi as a way to use up things growing in the garden we have a sudden glut of. Kim-Chard has become a necessity at certain times of the year and what about all those bloody courgettes that won't stop growing? KimChi them!

Fermentation Considerations

Why submerge the vegetables?

Ok let's look at what's going on "under the microscope". Let's take Sauerkraut as an example but all of this applies to Kimchi too. On the surface of the cabbage leaves there are different types of naturally occurring microbes. Here, we are interested in only one of these microbes called Lactobacillus. We want the Lactobacillus to multiply and to eat the sugars in the cabbage because when they do this, they produce lactic acid, which is why they are also called Lactic Acid Bacteria or LAB for short. Lactic acid is a natural preservative and helps create an intolerable environment for other microbes and pathogens that we DO NOT want in our sauerkraut.

So how do we get Lactobacillus to multiply then? We need to set up the right conditions. Lactobacillus thrive in an oxygen free environments e.g. when they are submerged in a brine. Salting the cabbage, draws water out of it which mixes with the salt to form a natural brine. Lactobacillus are also salt tolerant which other some microbes are not, so brine rather than water helps us cut out more harmful bacteria.

So the reason we weigh down the cabbage in the jar is because anything above the brine line, exposed to air, is more susceptible to contamination from other unwanted bacteria. If you give Lactobacillus a good chance to thrive, they will outcompete and kill off harmful microbes.

To weight it down, you can use something as simple as the outer leaf of a cabbage, pressed into the top of the jar. Whilst this leaf might be above the brine line in places that's totally fine as it can just be discarded once the sauerkraut has reached maturity and the rest of the submerged cabbage will be uncontaminated. You can buy fancy glass fermentation weights online although I have never used these and find that simply using another smaller jar or the cabbage leaf method works just fine. However if you want to experiment with these or other types of fermentation vessels you should.

Whatever you end up doing, don't worry too much. Without a weight, you can always just press the fermenting vegetables down with fork once a day, I've done this before and it's been fine.

Note on jars

There's a whole range of fermentation-specific jars available to buy, but to start with I'd suggest using store-bought jars that previously held things like olives or pickles. These standard jars can be saved after use, they're cheap and they seal while still releasing gases (see **Gases and burping**). You probably have some in your cupboards right now, so that's my suggested starting point. More expensive mason jars with screw tops or clip lids are also a great option, if a little pricier, just bear in mind that some of these will need burping from time to time.

Gases and burping

Lactobacillus produce lactic acid as a by-product of consuming the sugars in the vegetables, as discussed before but they also produce carbon dioxide (CO_2). This CO_2 gives a ferment a slightly fizzy sensation in the first few weeks, which fades over time. It's also essential for preservation, since it pushes oxygen, which Lactobacillus don't like, out of the jar. So if your jar is completely sealed tight, pressure will build up, and if not released there is the potential for an explosion. Luckily I have never had an exploding vessel but it is possible, so if you think you might miss burping the jar, something that lets gas out naturally is the way to go. Especially for Kimchi which I have found fermentation can often be vigorous. The first time I ever made Kimchi was in a sealed kilner jar, when I checked it after 4 days it burst violently open and sprayed kimchi all over my kitchen (and face)

Temperature

Lactobacillus need three things to multiply: food, an oxygen-free environment, and the right temperature. We've already covered the first two, the vegetable itself provides the food, and keeping it submerged under its own brine provides the oxygen-free environment, so temperature is the last piece of the puzzle.

Luckily, Lactobacillus can reproduce anywhere from 10ºC – 40ºC, so room temperature is totally fine for a tasty, safe ferment without any special adjustments. However as temperature rises, Lactobacillus multiply faster, so fermentation develops more quickly. In summer you might have a flavourful, sour Sauerkraut in just a week; in winter it might take 4 weeks.

Spillage

As fermentation gets going and CO_2 builds up, sometimes juices will escape the seal of the lid and spill down the side of the jar, especially if the jar is full and fermentation is vigorous. This is totally normal. I usually just put a saucer under the jar; you might need to clean it daily at the start, but after a while fermentation calms down and it stops being an issue.

Serving

Sauerkraut is, in my opinion, one of the most versatile vegetable ferments, it sits equally well in a salad, a cheese sandwich or as a side to roast pork. Most days it gets pulled straight from the fridge and served as a side with whatever we're eating.

- **As a sandwich pickle**, works brilliantly with cured or cooked meats, or cheese, or both! Add a little mustard.
- **In a salad**, adds crunch and zing to brighten things up. I especially like red cabbage kraut for colour.
- **As a side to umami-rich dishes**, goes well with pan-fried duck breast, beer-cooked sausages, marinated tempeh or sautéed mushrooms.

Basically, try it with everything and you'll see what you like it with.

Kimchi is much more of a flavour explosion than sauerkraut and can easily overwhelm more delicate flavours. However it can work brilliantly as a side to bolder, spicy flavours. Here are some of my favourites:

- **Kimchi & Cheese Toasty.** This is absolutely divine especially when paired with a mature cheddar on sourdough bread. Really melt the cheese and get a bit of charring on the bread, so delicious.
- **Kimchi & fried rice**, amp up the soy sauce and add a fried egg to the top. I'm getting hungry writing this haha.
- **With noodle dishes**, especially ramen/soup noodle dishes to add some spice and acid to cut through the umami flavours.

Troubleshooting

Occasionally contaminations can arise in your ferments. Most of the time they're perfectly fine and not harmful, but very occasionally you might want to throw a ferment away. Follow the salt ratios above and keep the vegetables submerged below the brine and you should be fine, but here's what to do if not:

White film on the surface

Might look slightly dusty or chalky and is in a very thin layer. This is called Kahm yeast and is naturally occurring and non-toxic. Simply scrape it off and eat the ferment underneath.

Coloured or fuzzy mould

More than a film, looks almost hairy or furry. This is an actual mould infection and the batch should be thrown away. It's the same kind of mould you might see on bread, and results from contamination, unclean equipment, the vegetables being exposed to air too much, or not enough salt.

Ferment is soft or mushy

Usually down to too little salt (salt helps preserve the crunch). It could also be too high a temperature, though it would have to be really hot, or it might simply be over-fermented and should have been moved to the fridge earlier. It’s not harmful but it’s not that pleasant to eat.